

Arrival choices

Previous lesson reminder:

Start with 1 and 2. Try 3 and 4 when ready. Reading and exact scribing are available.

1 Which response directly addresses tsunami exposure?

Support: Use the reminder strip or talk it through with an adult.

My response

2 Which meaning fits tectonic?

Support: Choose: Related to movement of Earth's plates OR People or things in harm's way.

My response

3 What was recorded at Coningsby on 19 July 2022?

Support: Use the model evidence anchor C or read one relevant card together.

My response

4 Why can the same heatwave create unequal risk?

Support: Give a reason when ready.

My response

Stretch: use a precise example or explain which detail supports your answer.

Evidence cards

Read, point or listen. Keep the source label with any detail you use.

A Japan 11 March 2011

A magnitude 9.1 earthquake occurred off north-east Honshu, Japan. Sudden movement on the subduction boundary displaced the seafloor and produced a tsunami.

Source: USGS event record, official20110311054624120_30

Limit: Magnitude measures the earthquake; it is not a direct count of people affected.

B Effects and responses

Shaking and coastal flooding damaged places and disrupted people's lives. Earthquake-resistant design addresses shaking; warnings, evacuation routes and higher ground address tsunami exposure.

Source: Authored explanation of the case and risk-management principles

Limit: A sea wall or warning cannot eliminate all risk. Do not judge people without considering access and warning time.

C Heat and unequal risk

On 19 July 2022 the Met Office recorded 40.3°C at Coningsby, 40.2°C at St James's Park and 39.6°C at Topcliffe. Outdoor work increases exposure; difficulty accessing cool spaces can increase vulnerability. Shade and accessible cool spaces can reduce harm.

Source: Met Office July 2022 report, station table; risk explanation authored for teaching

Limit: This explains risk factors, not a medical assessment of an individual.

D Reading a hazard feed

A USGS event row can include UTC time, place, magnitude and depth in kilometres. A short feed is a dated snapshot. It cannot predict the next earthquake.

Source: USGS earthquake feed documentation

Limit: Small samples, reporting thresholds and later revisions affect patterns.

Visual resource

Use this visual with the evidence cards and enquiry questions.



Source: Natural Earth public-domain outlines, labels simplified. Terms of use:
<https://www.naturalearthdata.com/about/terms-of-use/> (recorded in Sources and checks).

A detail I noticed and what it helps me explain

Data for the enquiry

Selected station maxima on 19 July 2022

Station	Maximum °C
Coningsby	40.3
St James's Park	40.2
Topcliffe	39.6
Sheffield	39.4

Units: °C

Source: Met Office July 2022 heatwave report, table on page 8

Limit: Four selected station maxima are not a national average or a long-term climate trend. The report marked values provisional at publication.

My labelled graph or data notes

Shared investigation

1. Read the July 2022 heatwave map and station data.
2. Separate weather event, exposure and vulnerability.

Work with a partner or adult, or use your own table. One person suggests; the other checks the evidence. Swap after one turn. Passing is allowed.

Move or match the cards

Cards to use
No access to a cool space
Working outdoors
High air temperature

Possible labels: Exposure / Hazard event / Vulnerability

Our explanation

Support: Use the glossary and one chunk of evidence. Plan the claim and supporting detail before explaining.

Further challenge: Evaluate why a station maximum does not measure every person's heat exposure.

Supported independent task

Complete this route only. Investigate extreme-weather hazards and risk.

1. Explain the heatwave's effects using the case evidence.
2. Identify exposure and vulnerability separately.
3. Evaluate one management response and explain which risk factor it addresses.

Support: Use the glossary and one chunk of evidence. Plan the claim and supporting detail before explaining. Complete the organiser one space at a time. The event was _____. Exposure includes _____. Vulnerability includes _____. The response reduces _____ by _____.

My choice or example

A detail from the card story or object

My reason

My check or next step

An adult may read or scribe your exact response. They should not choose the answer.

Standard independent task

Complete this route only. Investigate extreme-weather hazards and risk.

1. Explain the heatwave's effects using the case evidence.
2. Identify exposure and vulnerability separately.
3. Evaluate one management response and explain which risk factor it addresses.

Support: The event was _____. Exposure includes _____. Vulnerability includes _____. The response reduces _____ by _____.

My choice or example

A detail from the card story or object

My reason

My check or next step

Check the required evidence and revise one detail.

Stretch independent task

Complete this route only. Investigate extreme-weather hazards and risk.

1. Explain the heatwave's effects using the case evidence.
2. Identify exposure and vulnerability separately.
3. Evaluate one management response and explain which risk factor it addresses.
4. Evaluate why a station maximum does not measure every person's heat exposure.

Support: Choose the evidence independently. Use the frame only if it helps.

My choice or example

A detail from the card story or object

My reason

My check or next step

Evaluate why a station maximum does not measure every person's heat exposure.

Exit ticket

Choose a response mode. You may give your answers privately to an adult.

What was recorded at Coningsby on 19 July 2022?

My response

Why can the same heatwave create unequal risk?

My response

Support: The event was _____. Exposure includes _____. Vulnerability includes _____. The response reduces _____ by _____.

Further challenge: Evaluate why a station maximum does not measure every person's heat exposure.

One thing I changed or want to check
